

Main Research Idea: Probing the truth vector of advanced reasoning models through analyzing internal activations based on logic puzzle statements

Specific Process:

- This process is inspired by the 2023 CCS paper (<https://arxiv.org/pdf/2212.03827>) but with major differences; most of which, this process is supervised while still being scalable
- Instead of datasets of negational statements $\{(x_i^+, x_i^-)\}_{i=1}^n$, construct datasets in the following way
- Generate a logic equation $\Phi: \{\top, \perp\}^2 \rightarrow \{\top, \perp\}$ such that $\Phi(x, y) = 1$ only has one solution (x, y) with the solution satisfying $\neg(x \Leftrightarrow y)$
 - Example: $\Phi(x, y) = (x \Rightarrow y) \wedge (x \Leftrightarrow \neg y)$ giving one solution $(x, y) = (\perp, \top)$
- Use a language model to turn this to a logical puzzle P
 - Example of P
 $P = \text{"\"}$
Two people, N_1 and N_2 , each make a statement:
 N_1 : If my statement is true, then Bob's statement is true."
 N_2 : My statement has the opposite truth value from Alice's statement.
"\"
- **Randomize the names of N_1, N_2 and add distractions to the text**
- Give the following to the model
 - $x_i^{(1)} = P + "N_1 \text{ is telling the truth}"$
 - $x_i^{(2)} = P + "N_2 \text{ is telling the truth}"$
- Feed the statements to the reasoning model and find the activations $\phi(x_i^{(1)})$, $\phi(x_i^{(2)})$
- Option 1
 - Normalize the activations $\bar{\phi}(x_i^{(1)}) = \frac{\phi(x_i^{(1)}) - \mu^{(1)}}{\sigma^{(1)}}; \bar{\phi}(x_i^{(2)}) = \frac{\phi(x_i^{(2)}) - \mu^{(2)}}{\sigma^{(2)}}$
 - Run CCM
- Option 2
 - Since we're doing supervised learning, we can train the probe to find the truth directly
- Test the two options for whether the linear probe is able to learn anything

Benefits/Impacts: Addresses the issues with how CCS is currently done

- CCS is not scalable since it relies on human generated datasets, but this process is automatic
 - Addresses <https://arxiv.org/pdf/2501.16496#page=20&zoom=100,62,304>
- More importantly: Well-trained models more easily recognize "Paris is the capital of France" as what a knowledgeable human says based on their training on human text, but solving logic puzzles rely on actually knowing the truth
 - Many articles such as <https://arxiv.org/pdf/2312.10029> have pointed this out as a flaw with CCM
 - Addresses this issue as well
- Overall -> More invariant to semantic confounds

Possible Extension:

- For larger reasoning models, can you do the same with 3SAT problems?
 - Begin with a n -variable 3SAT problem with a unique solution
 - Example: A problem in (x, y, z) with solution $x = \top, y = \perp, z = \top$
 - Express the 3SAT problem in plain English
 - "Brian, Mike, and Chris all either all lie, all tell the truth; Brian: Mike and I do not both tell the truth; ..."